

Trading at Settlement (TAS) for Agricultural Futures

A FLEXIBLE AND TRANSPARENT WAY TO MANAGE SETTLEMENT PRICE UNCERTAINTY

Trading at Settlement (TAS) is an order type that allows a market participant to buy or sell futures contracts during the trading day equal to the yet-to-be determined settlement price, or at a price up to 4 ticks above or below that price.

Example:

For Corn, Soybean, SRW Wheat or HRW Wheat futures, the minimum price fluctuation or tick size is $\frac{1}{4}$ cent per bushel. A trader seeking to lock in the settlement price before it has been determined may enter any of the following TAS orders at any time during that trading day:

- A buyer may bid TAS 0 or a seller may offer TAS 0 to trade at the settlement price.
- A buyer may bid TAS +1, +2, +3 or +4 in an effort to incentivize the other side of the trade. Depending on the differential, these options would result in an assigned price equal to the settlement price plus \$.0025, \$.005, \$.0075, or \$.01 per bushel. The order book will show TAS ticks in the same units as the underlying futures.
- A seller may offer TAS -1, -2, -3 or -4 in an effort to incentivize the other side of the trade. Depending on the differential, these options would result in an assigned price equal to the settlement price minus \$.0025, \$.005, \$.0075 or \$.01 per bushel. The order book will show TAS ticks in the same units as the underlying futures.

Why TAS?

- Reduce uncertainty related to pricing around settlement
- Grain elevators and processors may use TAS orders to price forward contracts at or near the settlement value
- TAS orders offer a transparent alternative to floor based MOC orders which will no longer be available after July 2

Key Features

- Available on CME Globex
- Quotes will be published in real time on the screen throughout the trading day
- TAS will be available for outrights and spreads
- TAS order entry is not allowed prior to the beginning of each group's pre-open state, just like other order types
- TAS is traded in the same tick value as the underlying futures
- The matching algorithm will match that of the underlying futures

Which CBOT/CME Agricultural futures contracts are eligible for TAS?

	Contracts (Globex Code)	TAS Trading Hours	Contract Type	Contract Range	TAS Range
Grains & Oilseeds	Corn TAS (ZCT) Soybean TAS (SBT) Soybean Oil TAS (ZLT) Soybean Meal TAS (ZMT) Chicago SRW Wheat TAS (ZWT) KC HRW Wheat TAS (KET)	Sunday – Friday, 7:00 p.m. – 7:45 a.m. CT Monday – Friday, 8:30 a.m. – 1:15 p.m. CT	Outrights	First three listed futures contracts, plus the new crop month when not already represented in the first three	+/- 4 ticks
			Spreads	The nearby old crop-new crop spread First two consecutive calendar spreads	+/- 4 ticks
Livestock	Live Cattle TAS (LET) Feeder Cattle TAS (GFT) Lean Hogs TAS (HET)	Monday, 9:05 a.m. – 1 p.m. Tuesday – Friday, 8 a.m. – 1 p.m.	Outrights	First two listed futures contracts*	+/- 4 ticks
			Spreads	First consecutive calendar spread	+/- 4 ticks

* TAS is not offered on the May Lean Hog contract

What do values represent in the TAS Order Book?

	Corn, Wheat, Soybeans, Chicago SRW Wheat, KC HRW Wheat		Soybean Meal		Soybean Oil		Live Cattle, Lean Hogs, Feeder Cattle	
	Order Book	Value	Order Book	Value	Order Book	Value	Order Book	Value
TAS -4	-10	-\$.01	-4	-\$.40	-4	-\$.0004	-100	-\$.10
TAS -3	-6	-\$.0075	-3	-\$.30	-3	-\$.0003	-75	-\$.075
TAS -2	-4	-\$.005	-2	-\$.20	-2	-\$.0002	-50	-\$.050
TAS -1	-2	-\$.0025	-1	-\$.10	-1	-\$.0001	-25	-\$.025
TAS 0	0	0	0	0	0	0	0	0
TAS +1	2	\$.0025	1	\$.10	1	\$.0001	25	\$.025
TAS +2	4	\$.005	2	\$.20	2	\$.0002	50	\$.050
TAS +3	6	\$.0075	3	\$.30	3	\$.0003	75	\$.075
TAS +4	10	\$.01	4	\$.40	4	\$.0004	100	\$.10

What Would a TAS Order Book for Agricultural Futures Look Like?

Corn, Soybeans, Chicago SRW Wheat, KC HRW Wheat

- The number in the “Bid” and “Ofr” columns represent the differential from the settlement price in number of ticks. That may say -10, -6, -4, -2, 0, 2, 4, 6, 10.
- In this instance, TAS 0 or “TAS flat” is the best bid for the July contract, while TAS +1 (which is represented by the “2”) is the best offer for the same contract.

CC	Product	Description	Status	Qty	Bid	Ofr	Qty	Pin	Last	Change
- Corn TAS Future										
ZCT	Corn TAS Future	July15	Open	11,682	0 2	6,107			0	0
ZCT	Corn TAS Future	Sept15	Open	8,023	-2 0	1,087			+2	+2
ZCT	Corn TAS Future	Dec15	Open	4,683	-2 2	4,683			0	0
ZCT	Corn TAS Future	July15/Sept15	Open	7,249	-2 0	1,794			0	0
ZCT	Corn TAS Future	July15/Dec15	Open	284	-4 0	4,284			-2	-2

Soybean Meal, Soybean Oil

- The number in the “Bid” and “Ofr” columns represent the differential from the settlement price in number of ticks. That may say -4, -3, -2, -1, 0, 1, 2, 3 or 4.
- In this instance, TAS -1 (which is represented by the “-1”) is the best bid for the July contract, while TAS +1 (which is represented by the “1”) is the best offer for the same contract.

CC	Product	Description	Status	Qty	Bid	Ofr	Qty	Pin	Last	Change
Soybean Meal TAS Future										
ZMT	Soybean Meal TAS Future	July15	Open	8,478	-1	1	7,764		0	0
ZMT	Soybean Meal TAS Future	Aug15	Open	5,772	0	1	1,428		1	+1
ZMT	Soybean Meal TAS Future	Sept15	Open	3,860	0	2	3,651		0	0
ZMT	Soybean Meal TAS Future	July15/Aug15	Open	2,439	-1	1	1,635		0	0

Live Cattle, Lean Hogs, Feeder Cattle

- The number in the “Bid” and “Ofr” columns represent the differential from the settlement price in number of ticks. That may say -100, -75, -50, -25, 0, 25, 50, 75, 100.
- In this instance, TAS -1 (which is represented by the “-25”) is the best bid for the June contract, while TAS 0 or “TAS flat” is the best offer for the same contract.

CC	Product	Description	Status	Qty	Bid	Ofr	Qty	Pin	Last	Change
Live Cattle TAS Future										
IFT	Live Cattle TAS Future	June15	Open	3,673	-25	0	2,658		25	+25
LET	Live Cattle TAS Future	Aug15	Open	2,578	-50	0	1,044		-25	0
LET	Live Cattle TAS Future	June15/Aug15	Open	1,569	-25	25	578		0	0

TAS on Outrights:

Grain and Oilseeds Example:

A producer wants to sell five new crop futures contracts of December Corn at the settlement price.

- The producer offers to sell 5 contracts at the settlement price – “TAS 0” or “TAS flat”
- At the same time, a grain processor wants to hedge his grain inputs by purchasing 5 December contracts and prefers to receive at or near the settlement price, so he places an order to buy 5 December contracts at TAS flat

The trades are matched. Later that afternoon at 1:15, Dec corn settles at \$4.00. The resulting executed price to the buyer and seller is therefore \$4.00.

Livestock Example:

A beef producer wants to sell one April live cattle futures contract at the settlement price.

- There is no bid at “TAS flat”
- He enters a TAS-1 offer in the order book in effort to incentivize the other side of the trade
- Meanwhile, a beef packer wants to buy one April Live Cattle futures contract at or near the settlement price, so she lifts the offer and buys one contract at TAS-1, the settlement price minus one tick, or 2 ½ cents per hundredweight.

The trades are matched. Later that afternoon at 1:00, April Live Cattle settles at \$160.875 per hundredweight. The resulting executed price to the buyer and seller is therefore \$160.850 (settlement price minus \$.025)

TAS on Calendar Spreads

TAS may also be used in executing calendar spreads.

- If the TAS spread trades at a positive differential (i.e. TAS +1), the nearby leg of the spread will be priced at the settlement price for that contract month plus the applicable TAS differential. The deferred leg will be priced at the settlement price for that contract month.
- If the TAS spread trades at a negative differential (i.e. TAS -1), the nearby leg of the spread will be priced at the settlement price for that contract month. The deferred leg will be priced at the settlement price for that contract month plus the absolute value of the applicable TAS differential.

Example:

A November-January soybean calendar spread trades at TAS +1. Later that afternoon, the November contract settles at \$11.00 and the January contract settles at \$11.08

- The November leg will be priced at \$11.0025, which is the settlement price of \$11.00 plus the TAS price increment of +1 (\$.0025)
- The January leg will be priced at the contract settlement price of \$11.08
- As a result, the TAS spread is $-\$.0775$ versus $-\$.08$ for the standard spread settlement

Example:

A November-January soybean calendar spread trades at TAS -1. Later that afternoon, the November contract settles at \$11.00 and the January contract settles at \$11.08

- The November leg will be priced at the contract November settlement price of \$11.00.
- The January leg will be priced at \$11.0825, which is the settlement price of \$11.08 plus the absolute value of the TAS differential of -1 ($-\$.0025$)
- As a result, the TAS spread is $-\$.0825$ versus $-\$.08$ for the standard spread settlement

TAS Protocol on Limit or Near-Limit Daily Price Moves

- TAS orders will be filled at +/- up to 4 ticks above or below the settlement price regardless of daily price movements.
- On limit or near-limit days, the price assigned could potentially be outside of hard limits by up to one cent per bushel, or 4 ticks, in grain and oilseed futures.
- Price assignments in livestock futures may be outside of hard limits by up to 10 cents per hundredweight, or 4 ticks.

Why isn't TAS offered on further-deferred contracts, Rough Rice Futures, Oat Futures or Dairy Futures?

- In order to preserve the integrity of our markets, an extensive liquidity analysis was conducted in relation to supporting TAS. To prevent any possible adverse market impact, only contracts and spreads with appropriate liquidity will have TAS order capability.

For more information about TAS for Agricultural Futures, please contact agtas@cmegroup.com or visit cmegroup.com/agtas.

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